Hibernate inserts:

~~26,545 ms~~

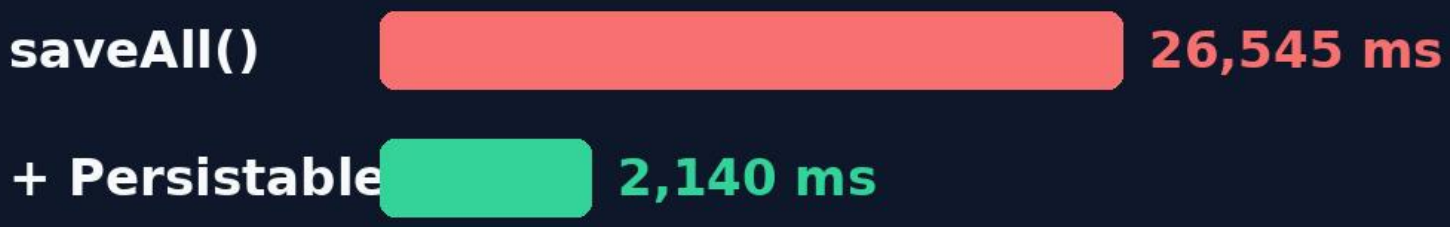
646 ms

3 settings close ~90% of the gap to raw Bulk Copy — 100k inserts, median of 10 runs

swipe →

1. Kill the pre-INSERT SELECTs

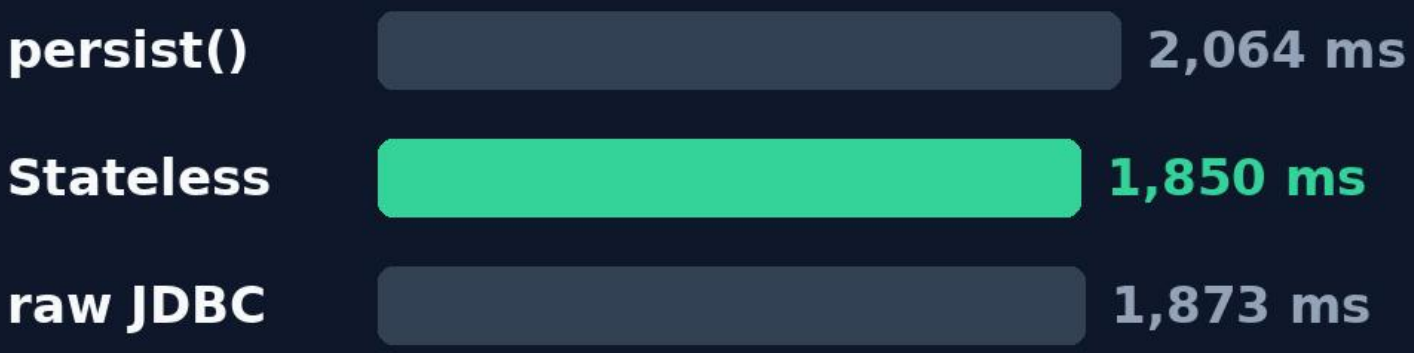
saveAll() calls merge() — a SELECT before every INSERT (100k extra)



isNew() = true → ~12x faster. 10 lines of code.

2. StatelessSession = bare metal

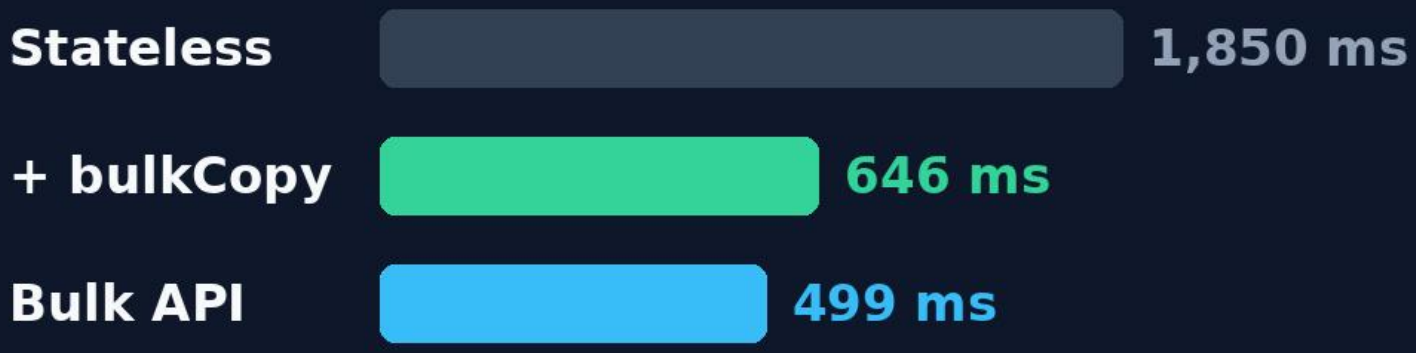
Drops L1 cache, dirty checking, cascading — constant memory, no entity tracking



Within ~1% of raw JDBC.

3. Turn on the bulkCopy flag

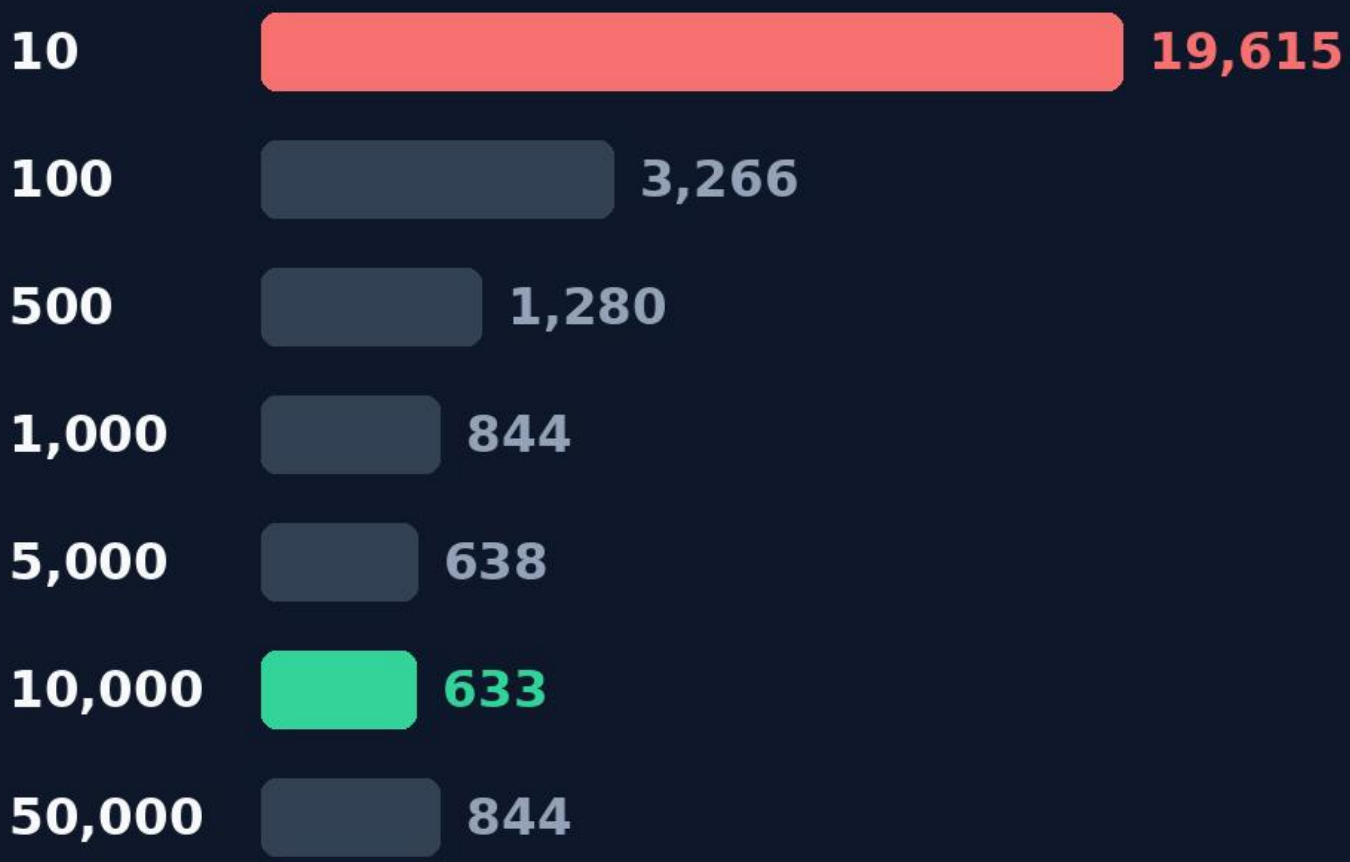
useBulkCopyForBatchInsert=true —
each executeBatch() = one Bulk Copy stream



2.9x — only ~30% off the raw Bulk API.

Then tune jdbc.batch_size

StatelessSession + bulkCopy, 100k inserts:



batch_size=1 fails outright — and 10 is the cliff: ~20 s, worse than not batching at all. Optimum: 5,000–10,000.

Full benchmarks + runnable code

professionaldev.dev

github.com/mmarko88/blog-examples

Batch-size sweep, Persistable source, StatelessSession repo — link in comments.